



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VIII Semester B.E EEE

Course Code & Title: EE8019 – Smart Grid

Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic:** Unit IV / Power Quality issues of Grid Connected Renewable Energy Sources

- **Course Outcome:** CO 4

- **Topic Learning Outcome:** TLO 10

- **Activity Chosen:** Ungraded Quiz

- **Justification:**

- The chosen topic deals with power quality issues present in the grid and its mitigation methods. Usually the students will find it difficult to understand the methods and also to properly address the power quality issues arises in the grid while interconnecting renewable energy sources to it. By conducting this ungraded quiz, I helped them to understand the concepts better.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

- The students are taken to the Power system simulation laboratory and each one is occupying the one system with Internet connectivity.
- The quiz has been created in the Quizizz online tool and pass code has been shared with the students.
- All the students have started the quiz at the same time. At the end of the quiz, the correct answers will be shown to them in the screen.
- Finally, I explained the correct answer for the quiz questions with proper reasons.

- **CO – PO / PSO mapping:**

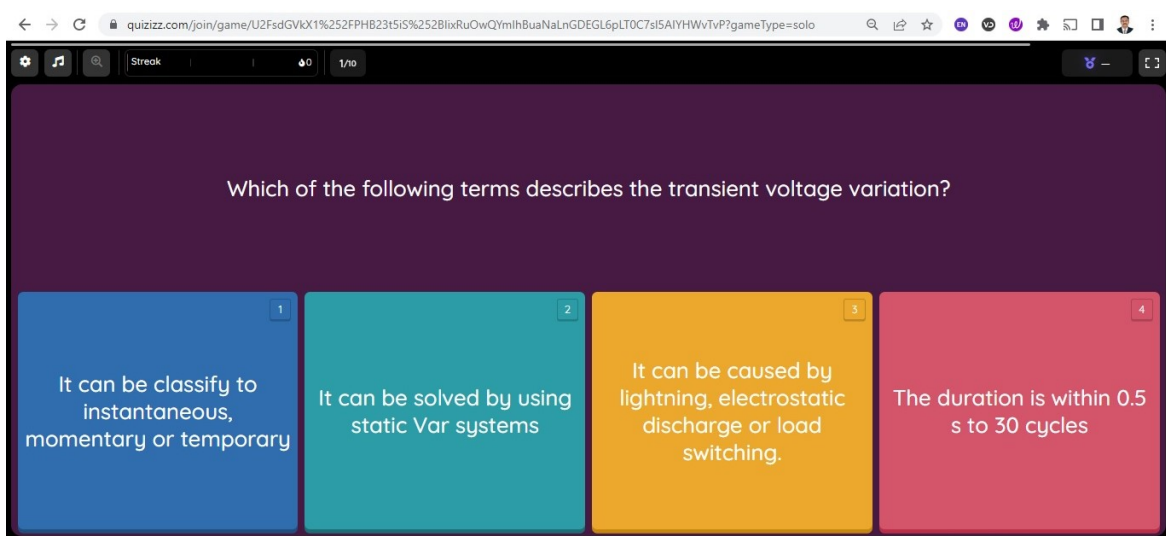
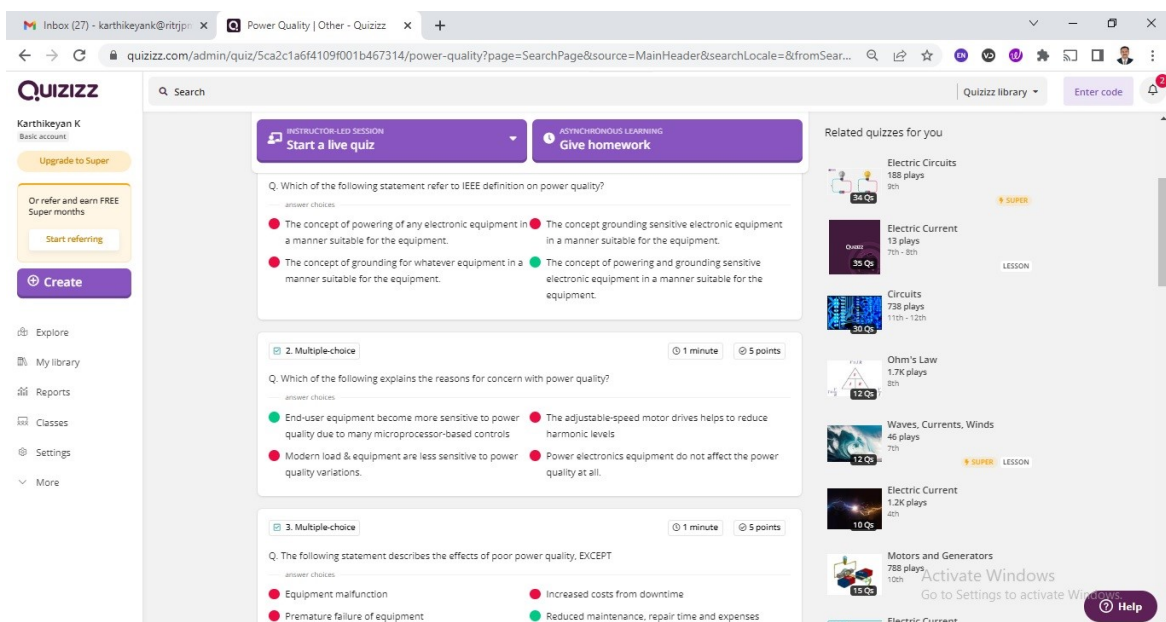
CO	PO 1	PO 2	PO 5	PO 8	PO 12	PSO 1
CO2	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 5	PO 8
	1	1
Justification for correlation	The students are using online tool for doing this quiz. Hence it is mapped with this PO at level 1	The students are doing this quiz without discussing and copying from others. Hence it is mapped with this PO at level 1

• **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ The students have enjoyed the quiz. They told they have been given with the chance of remembering the concepts in the same class itself.

- ❖ ***Benefit of the practice:***

- ✓ The students are getting the feel of involved in the learning process.
 - ✓ Since the quiz is not included in the marks calculation, the students have done the quiz without any test fear. Hence, they participated freely.

- ❖ ***Challenges faced in implementation:***

- ✓ Few students have not much interest in attending the quiz sincerely, since the quiz is ungraded. Managing those students and making them to do the quiz is difficult task for me in this activity.

References:

- ❖ I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, “Smart Grid Fundamentals & Applications”, New Age International Publishers, 1st Edition, 2019.
- ❖ Bharat Modi, Anu Prakash, and Yogesh Kumar, “Fundamentals of Smart Grid Technology”, S. K. Kataria & Sons, 1st Edition, 2019.
- ❖ <https://article.sciencepublishinggroup.com/html/10.11648.j.ijepe.s.2015040501.14.html>
- ❖ <https://www.icrepq.com/icrepq'10/505-Khadem.pdf>

Signature of Faculty Member

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Course Code & Title: EE8019 – Smart Grid

Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic:** Unit I / National Initiatives in Smart Grid
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen:** Case Study Report
- **Justification:**
 - The chosen topic will be of facts and figures about the pilot projects in India. The current status of the Smart Grid projects needs to be checked and presented. If all these data are presented in the conventional class room lecture, the students may not be interested and may not understand fully.
 - If the students are encouraged to present these data in any Info graphic form / table form, then they will understand the concepts better and also enjoy the session. That is the reason for choosing this activity “Case Study Report”.
- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**
 - The students are taken to the Power System Simulation Laboratory and they are provided with individual computer system with internet connectivity.
 - The components/technologies of the smart grid and the Smart Grid pilot projects of the India have been briefly explained to the students.
 - The students are asked to prepare the case study report on the topic Smart Grid Initiatives in India.
 - The guidelines for the contents of the report and formatting of the report have been told to the students.
 - At the end of the period, the students have submitted their report in the CANVAS – LMS.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 5	PO 12	PSO 1
CO2	3	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 5	PO 8	PO 10
	1	1	1
Justification for correlation	The students will be using the internet, drawing tools etc., for preparing the report.	The students will follow the ethics and they will not be allowed to copy from other students.	The students are writing the report; hence their written communication is improved.

• **Images / Screenshot of the practice:**

The screenshot shows a Canvas LMS interface. The left sidebar contains navigation links: Home, Announcements, Assignments, Discussions, Grades, People, Pages, Files, Syllabus, Outcomes, Rubrics, Quizzes, Modules, BigBlueButton, Collaborations, Attendance, and New Analytics. The main content area displays the assignment details for 'Report on Smart Grid Initiatives in India'. It includes a message to students asking for a four-page report on Smart Grid (SG) initiatives or pilot projects in India, due before 30.04.2022. The report should contain information about the name of the SG pilot projects, their essential features, and their current status. The assignment is worth 10 points and is submitted via file upload. A table at the bottom shows the due date (May 10), the audience (Everyone), the available time (Mar 8 at 12am), and the submission deadline (May 10 at 11:59pm). On the right, there are options to download or re-upload submissions, and a status indicating that 20 out of 22 submissions have been graded.

The screenshot shows a student's submission for the 'Report on Smart Grid Initiatives in India' assignment. The submission is titled 'SMART GRID PILOT PROJECTS IN INDIA'. The report content includes a definition of Smart Grid as an Electrical Grid with Automation, Communication and IT systems, and a list of 14 smart grid pilot projects shortlisted in India. The projects listed are: CESC (Karnataka) – AMI, outage management, peak load management, microgrid and distributed generation with an initial 21,800 consumers in the Mysore Additional City area; Andhra Pradesh CPDCL – AMI, outage management, peak load management and power quality management with 11,900 consumers in the Jeedimetla suburb of Hyderabad; Assam PDCL – AMI, outage management, peak load management, power quality management and distributed generation with 15,000 consumers in the Guwahati area; and Gujarat VCL – AMI, outage management, peak load management and power quality management with 39,400 consumers in Naroda and Deesa. The submission is graded 8.48 / 10 (85%) and has a word count of 825 words. The submission was made on Mar 29 at 11:51am. The right sidebar shows the submission details, including the word count, submitted files, and the assessment score.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ Many students requested to conduct this type of activity frequently, because it provided them an opportunity to have hands on training with MS-word and some drawing tool.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The students have prepared their report and hence they got better understanding about the concept, it was evident once I asked few to comprehend the points written in their report.
 - ✓ The student's presentation skills got improved since they prepared report and it was also useful for their exam preparation.

- ❖ ***Challenges faced in implementation:***

- ✓ Initially the students hesitated to prepare a report. I explained them to understand the work they need to complete and helped them in doing so in all the phases.
 - ✓ Managing the non performing students was quite difficult. They used to login some other websites and not doing the task.

References:

- ❖ I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, "Smart Grid Fundamentals & Applications", New Age International Publishers, 1st Edition, 2019.
- ❖ Bharat Modi, Anu Prakash, and Yogesh Kumar, "Fundamentals of Smart Grid Technology", S. K. Kataria & Sons, 1st Edition, 2019.
- ❖ <https://www.iitk.ac.in/npsc/Papers/NPSC2014/1569993451.pdf>
- ❖ https://indiasmartgrid.org/nsgm_about.php
- ❖ <https://events.development.asia/system/files/materials/2018/04/201804-smart-grid-developments-india.pdf>

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Innovative Practice Description

- **Unit / Topic:** Unit II / Plugin Hybrid Electric Vehicles (PHEV)
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 6
- **Activity Chosen:** Think – Pair - Share
- **Justification:**
 - The concept of Electric Vehicle is relatively a new concept for the students. Especially the operation of IC engine based vehicles completely new to the EEE students. The PHEV is a hybrid of IC engine and EV and hence the students may find it difficult to understand the concepts. Therefore, the team activity “Think-Pair-Share” has been planned for this topic.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
 - The students have been grouped into smaller groups of four members, with one bright student, two average learners and one slow learner in each group. One average learner from each group has been made as a recorder.
 - I have shown the power point presentation for the quick recap of the concepts like Vehicle types, benefits of Electric Vehicle, and the constructional difference between IC engine vehicle and Electric Vehicle.
 - The detailed operation procedure of the IC engine vehicle has been explained to the students.
 - The students have been told to recollect the concepts of Plug-in Hybrid Vehicle (PHEV) and then asked to discuss about it within the group for 5 minutes.
 - Finally, two recorders have been asked to share their results to the whole class through chalk and talk technique. All other students interacted with them for proper understanding the concepts.

• **CO – PO / PSO mapping:**

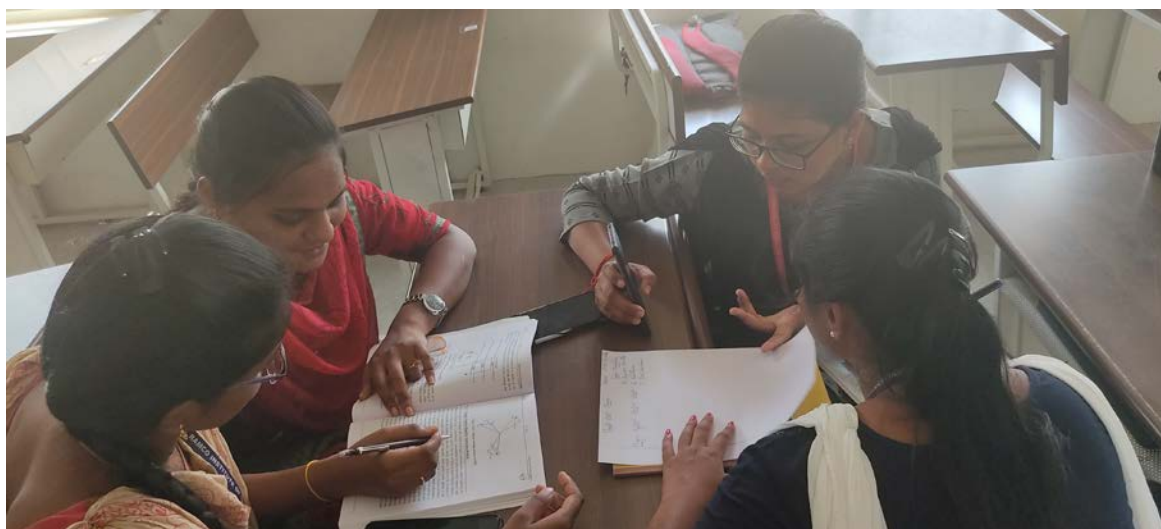
CO	PO 1	PO 2	PO 5	PO 8	PO12	PSO 1
CO2	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 9	PO 10
	1	1
Justification for correlation	The students will do this activity in team. Hence it is mapped with PO 9 with level 1.	The students will communicate with each other and do the oral presentation. Hence this activity is mapped with PO 10 with level 1.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ The students have enjoyed the activity. Most of the students requested me to conduct such activities in the classes. When I did this activity, they had a great confidence in learning the new concept.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The students have understood the concepts clearly and it has been evident from the points they have shared with the classroom.
- ✓ The students have presented the points to the class; it cleared any ambiguity in understanding the important points in the chosen topic.

❖ ***Challenges faced in implementation:***

- ✓ The activity has been planned for 15 minutes, but it has been completed after 16 minutes only. Next time, I should plan the activity with proper time.
- ✓ The group has to form as heterogeneous, and then only I can take care about the non-participating students.

References:

- ❖ I.S. Jha, Subir Sen, Rajesh Kumar, and D.P. Kothari, “Smart Grid Fundamentals & Applications”, New Age International Publishers, 1st Edition, 2019.
- ❖ Bharat Modi, Anu Prakash, and Yogesh Kumar, “Fundamentals of Smart Grid Technology”, S. K. Kataria & Sons, 1st Edition, 2019.
- ❖ https://afdc.energy.gov/vehicles/electric_basics_phev.html

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